

Final Project : User Manual

**FPGA Based Prime Number Guessing Game With ROM
Based Multi User Authentication And RAM-Score Tracking**

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Primer Number Guessing Game Description:

The system is an FPGA-based **multiuser prime number guessing game** designed using Verilog. The game allows different players to log in, select a difficulty level, play within a countdown timer, and track their scores individually and globally.

First, the user logs in through the **multiuser authentication system** using the password/ID input switches and the game start button. After successful authentication, the system turns on the logged-in status and allows the player to select a game level. The selected level controls two important parts of the game: the **target prime number range** and the **timer duration**.

The game has three difficulty levels:

Level 1:

Prime target range = 2 to 23

Timer duration = 099 seconds

Level 2:

Prime target range = 2 to 53

Timer duration = 160 seconds

Level 3:

Prime target range = 2 to 97

Timer duration = 220 seconds

Once the game starts, the **PrimeGameController FSM** controls the overall game flow. It reconfigures the timer, requests a new target prime number, clears the previous guess, shows the start indication, and then enters the active game state. The hidden target prime is generated by the **PrimeTargetGenerator**, which uses an LFSR-based random index and a ROM containing prime numbers.

The player enters a guess using the input switches and the player load button. The guess is entered as two digits, so the first load stores the tens digit and the second load stores the ones digit. After both digits are entered, the system forms the complete guessed number and sends it to the game controller and prime checker.

The system then checks the guess and gives feedback using the hint display:

F → The entered number is not prime / invalid guess

S → The guess is smaller than the target prime

L → The guess is larger than the target prime

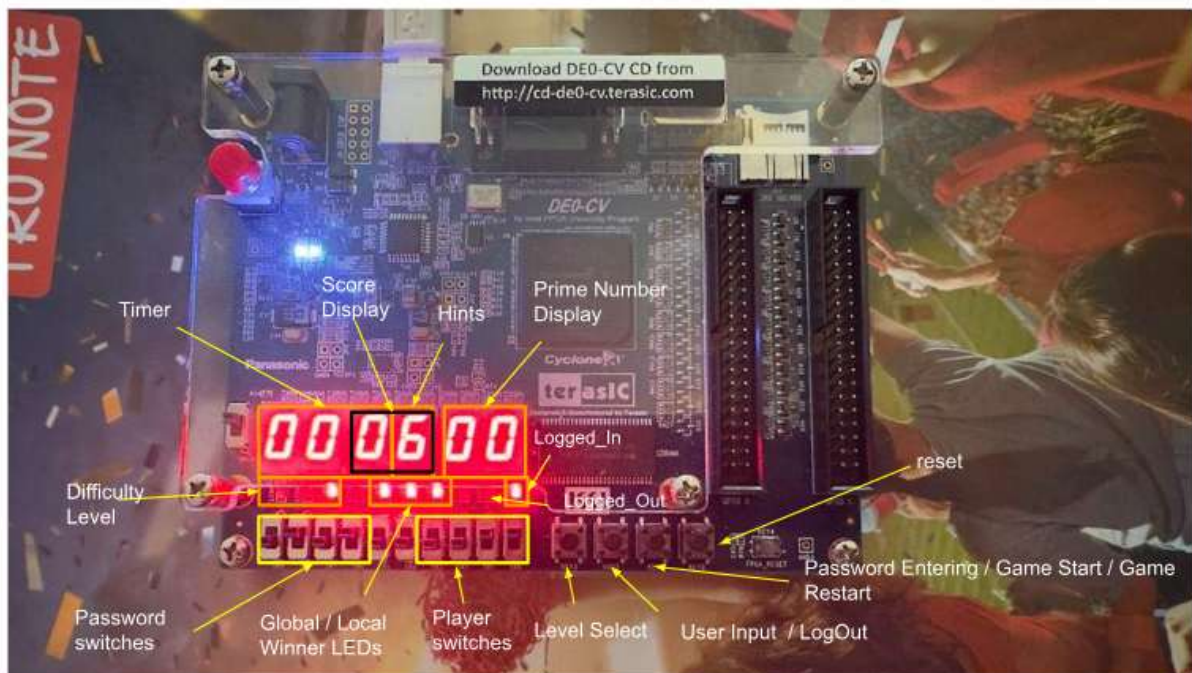
C → The guess is correct

If the player guesses correctly, the score is incremented, the current guess is cleared back to 00, and a new target prime number is generated silently. The player continues guessing until the timer reaches zero.

When the timer times out, the game enters the game-over state and displays the final score. The **ScoreTracking** section compares the final score with the player's previous personal best and the overall global best score. If the score is the player's first score or higher than their previous personal best, the system marks it as a **personal/local winner**. If the score is the first score overall or higher than the previous global best, the system marks it as a **global winner**.

The LEDs are used to show login/logout status, selected level, and winner indication, while the seven-segment displays show the timer, score, hints, and the player's entered guess.

Board picture illustrating FPGA Components usage:



c) Step-by-step description of authentication and gameplay

Authentication Steps

1. Reset the system

The player first resets the system. This clears the previous game status, timer, score, guess input, and authentication progress.

2. Enter the 4-digit Player ID

The player uses Password_Switch[9:6] to enter the Player ID one digit at a time. After setting each digit on the switches, the player presses the authentication/game start button to load it.

ID digit 1 → press enter

ID digit 2 → press enter

ID digit 3 → press enter

ID digit 4 → press enter

3. ID checking

After four digits are entered, the ID_Checking_Module compares the entered ID with the stored IDs in PlayerID_ROM. If the ID matches, the system identifies the player.

4. Enter the 6-digit password

After the ID is accepted, the player enters the password using the same switches. Each password digit is loaded by pressing the authentication/game start button.

Password digit 1 → press enter
Password digit 2 → press enter
Password digit 3 → press enter
Password digit 4 → press enter
Password digit 5 → press enter
Password digit 6 → press enter

5. **Password checking**

The Password_Checking_Module compares the entered password with the stored password in Password_ROM for that player.

6. **Login successful**

If both the ID and password are correct, the system asserts Logged_In, turns on the logged-in LED, and allows the player to start the game. If authentication fails, the player must enter valid credentials again.

Gameplay Steps

1. **Select the level**

After login, the player selects a difficulty level using the level button. The selected level controls both the target prime range and the timer duration.

Level 1 → Prime range: 2 to 23 → Timer: 099 seconds

Level 2 → Prime range: 2 to 53 → Timer: 160 seconds

Level 3 → Prime range: 2 to 97 → Timer: 220 seconds

2. **Start the game**

The player presses the game start button. The game controller reconfigures the timer, clears the previous guess, resets the game status, and requests a new hidden target prime number.

3. **Target prime generation**

The target prime is generated internally using the LFSR-based random index generator and the prime number ROM. The target is hidden from the player.

4. **Timer starts counting down**

The three-digit timer starts from the selected level time. During the game, the timer is shown on the timer displays.

5. **Enter a two-digit guess**

The player uses Player_Switch[3:0] and the Player_LoadButton to enter the guess digit by digit.

First load → stores the tens digit

Second load → stores the ones digit

Example: to enter 53, set the switches to 5 and press load, then set the switches to 3 and press load.

6. **Guess checking**

After two digits are entered, the system forms the complete guess number. The

PrimeChecker checks whether the guess is prime, and the game controller compares it with the hidden target prime.

7. **Hint display**

The system displays a hint based on the guess result.

F → entered number is not prime / invalid

S → guess is smaller than the target

L → guess is larger than the target

C → guess is correct

8. **Correct guess**

If the guess is correct, the score increases by one. The entered guess is cleared back to 00, and a new hidden target prime is generated while the timer continues.

9. **Game over**

When the timer reaches 000, the game stops and the final score is displayed on the score displays.

10. **Winner checking**

After game over, the score tracking module compares the final score with the player's personal best and the global best score. If the score is a new personal best or global best, the corresponding winner indication is activated.

11. **Logout**

After the game ends, the player can press the player load/logout button to log out and return the system to the authentication stage.